

Sutram SRM

Deep Learning Super-Resolution Mapping from Medium-Resolution Satellite Imagery — Sentinel-2 10 m → 2.5 m with per-pixel trust.

Team **Sutram** · Smart India Hackathon · Full project documentation · 25 August 2026 · 3,100 lines of Python across 27 modules

0.917

field-boundary F1
(bicubic 0.800)

1.808°

spectral angle
best of all branches

0.00 m

geolocation drift
every product

1st

on every metric
vs bicubic & SEN2SR

The system in one paragraph

Sentinel-2 images all of India every five days, free, at 10 m per pixel — enough to see a field, not its boundary. Sutram SRM reconstructs that imagery at 2.5 m using deep learning, and ships a per-pixel confidence map alongside it so an analyst always knows which sharpened detail is supported by sensor data and which the model inferred. Trained on real Sentinel-2/VENμS pairs over **Telangana** and four further biomes, it improves field-boundary delineation from F1 0.800 to **0.917** — not merely sharper pixels, but measurably better analysis.

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1. Problem and approach

1.1 The operational problem

Medium-resolution imagery (10–30 m) underpins operational Earth observation in India: free, national coverage, five-day revisit. But at 10 m a village road is one pixel wide, a small building is sub-pixel, and a field boundary is a blur. Analysts face a hard choice:

Option	Cost	Consequence
Free 10 m Sentinel-2	Free, national, 5-day revisit	Cannot resolve the features the decision depends on
Commercial sub-metre	Thousands of rupees per scene	Unaffordable at state scale; infrequent revisit

The problem we actually solve

Generative super-resolution offers a third option — and introduces a new risk. A model that sharpens imagery is a model that can *invent* imagery. An analyst shown a crisp building the sensor never observed may act on a fabricated pixel. **That risk, not image quality, is why super-resolution is not already operational in most agencies.** Several open models already sharpen Sentinel-2 well. The unsolved problem is making the result *trustworthy enough to act on*.

1.2 Design response

- **Multiple models, one interface.** Four branches — a baseline, a fidelity-constrained CNN, a generative diffusion model, and our own trained network — run behind a single `predict()` call, so they can be compared honestly.
- **Uncertainty travels inside the product.** Confidence is a band in the same GeoTIFF as the imagery; it cannot be separated from the pixels it qualifies.
- **Physical validation, not just statistical.** Every product is checked by re-observing it through the sensor's own point-spread function.
- **Utility over aesthetics.** Success is measured by whether downstream analysis improves, not by how sharp the picture looks.

2. System architecture

```
Sentinel-2 L2A (B04 B03 B02 B08 @ 10 m + SCL)
|
[1] Preprocessing ..... SCL cloud mask -> reflectance scale -> tile 128px/32 overlap
|
[2] Branches (common interface, tiling internal to each)
    |-- Bicubic ..... baseline control
    |-- SEN2SR ..... fidelity, hard radiometric constraint (ESA OpenSR)
    |-- LDSR-S2 ..... generative detail + per-pixel sigma from N samples
    |-- Ours ..... SPAN CNN trained on SEN2VENuS / KUDALIAR (Telangana)
    |
[3] Trust layer ..... LR-consistency | spectral angle | branch spread | sigma
    |                                     -> fused per-pixel confidence in [0,1]
[4] Validation ..... Wald protocol | consistency check | calibration curve
    |
[5] Product ..... 2.5 m COG GeoTIFF: 4 SR bands + sigma + confidence
    |
[6] Downstream ..... field boundaries | urban structure | change detection
```

Module	Responsibility
src/srm/preprocess/	SCL masking, reflectance scaling, tile geometry, Hann blend weights
src/srm/models/	Branch wrappers behind one interface; feathered tiled inference
src/srm/trust/	PSF downsampling, LR-consistency, spectral angle, Δ NDVI, confidence fusion
src/srm/validate/	Wald protocol, consistency check, calibration curve, downstream tasks
src/srm/metrics/	Reflectance-aware PSNR / SSIM / SAM / ERGAS
src/srm/train/	Composite loss, shard dataset, model construction
src/srm/io/	COG writing, CRS and footprint preservation and verification
src/srm/data/	SEN2VENuS nested-archive pair loader, $\times 2/\times 4$ pairing modes

2.1 Technology choices

Layer	Choice	Rationale
Language / ML	Python 3.12, PyTorch 2.13	Geospatial + ML ecosystem; MPS support makes local training viable
Geospatial I/O	rasterio / rioxarray (GDAL)	Correct CRS and affine handling; native COG writing
Fidelity model	SEN2SR — SPAN CNN, 472k params	Hard radiometric constraint built in; CC0; peer-reviewed (RSE 2025)
Generative model	LDSR-S2 latent diffusion, 169M params	Native per-pixel uncertainty through stochastic sampling
Our model	SPAN CNN, warm-started	Converges in minutes on free compute; a from-scratch transformer would not converge in schedule
Interface	Streamlit (two apps)	Viewer is a viewer; a custom SPA would consume the budget for no capability

3. Data

3.1 Training data — SEN2VEN μ S v2.0.0

Real Sentinel-2 $\leftrightarrow$ VEN μ S pairs acquired the **same day** over the same ground, already co-registered.

Why same-day pairing is non-negotiable

Puri & Kotze (AGILE 2022) traced invented objects in their SRGAN outputs to a 25–30 day gap between low- and high-resolution acquisition: the model learned to reconstruct *change* as if it were *detail*. Same-day pairs eliminate that failure mode at source rather than patching it downstream.

Site	Location	Patches	Role
KUDALIAR	Telangana, India (tile 44QKE)	7,269	Primary — puts real Indian terrain in training
ANJI	China	2,312	Mixed agriculture / forest
MAD-AMBO	Madagascar	1,442	Tropical
SUDOUÉ-4	France	935	Agricultural parcels
ESTUAMAR	Spain	911	Coastal / estuary

Two packed datasets exist: **KUDALIAR-only** (5,400 train / 600 val) and **multi-site** (8,640 / 960 across all five sites). Both were trained; the multi-site model is the one reported throughout this document, on the evidence in §7.6.

3.2 The scale subtlety

SEN2VEN μ S is natively $\times 2$, not $\times 4$

Pairs are 10 m Sentinel-2 against 5 m VEN μ S (128 $\rightarrow$ 256 px). Our deployment target is $\times 4$ (10 m $\rightarrow$ 2.5 m), and no open global 2.5 m reference exists. Stating this openly is stronger than letting a reviewer discover it.

Mode	Input	Target	Scale
--scale 2	real S2 10 m	real VEN μ S 5 m	$\times 2$, fully real
--scale 4 (used)	S2 degraded to 20 m <i>via the S2 PSF</i>	real VENμS 5 m	$\times 4$, real HR target

Only the *input* is synthetic, and it is synthesised with the sensor's own response rather than bicubic — a model trained to invert bicubic learns the wrong operator. This is Wald's protocol applied to training rather than evaluation, and it is what makes a $\times 4$ claim defensible without owning 2.5 m imagery.

3.3 Validation data — real Sentinel-2 over Telangana

Two dates fetched anonymously from the AWS public Sentinel-2 archive via Earth Search STAC (no Copernicus account required), from **the same MGRS tile as the training data**:

Scene	Date	Cloud	Size	Purpose
S2B_44QKE_20251113	13 Nov 2025	0.0%	1289 $\times$ 1344 @10 m	Rabi sowing — change-detection reference
S2A_44QKE_20260310	10 Mar 2026	0.0%	1289 $\times$ 1344 @10 m	Pre-harvest — primary validation scene

3.4 Bands

4 in, 6 out. Input is the four natively-10 m bands in model order **B04, B03, B02, B08** (red, green, blue, NIR) — order verified from the model's own metadata, and not alphabetical. NIR is included deliberately: without it NDVI is impossible and every crop application collapses. Output adds `sigma` and `confidence`. The 20 m red-edge/SWIR and 60 m atmospheric bands are out of scope for v1.

4. Model and training

4.1 Architecture

SPAN CNN (from ESA's SEN2SR Lite release), warm-started from the released weights and fine-tuned on SEN2VEN μ S. Two variants exist: **lite** (24 channels, 6 blocks, 572k params — the model reported here) and **wide** (48 channels, 10 blocks, 3.63M params).

Why reuse an architecture rather than invent one

On free-tier compute, warm-starting from pretrained weights reaches a useful model in minutes; a transformer trained from scratch would not converge inside the schedule at all. What makes the result ours is the *training data* (real Indian terrain) and the *objective* (physical consistency + spectral angle) — not a novel block diagram. Claiming architectural novelty we do not have would be the weaker position.

4.2 Loss function

Deliberately not the standard ESRGAN recipe. VGG-19 perceptual loss is trained on 8-bit consumer photographs and mismatched to 16-bit reflectance; plain adversarial training cost Puri & Kotze roughly 0.2 SSIM in artefacts. We optimise what the problem statement actually asks for:

Term	Weight	Purpose
L1 on reflectance	1.0	Radiometric accuracy
LR-consistency through the PSF	0.5	Do not invent radiometry
Spectral angle (SAM)	0.1	Spectral consistency, brightness-invariant
Gradient	0.1	Sharpness without a photographic prior

The consistency term is the interesting one

It is the differentiable form of the check the trust layer runs at inference. The model is penalised *during training* for exactly the behaviour we flag at *deployment* — objective and safety check measure the same physical quantity. This is why our model leads every consistency metric on real imagery.

Augmentation is dihedral only (rotations, flips). No colour jitter: shifting reflectance would teach the model that radiometry is negotiable, the one thing this pipeline must not learn.

4.3 Training run

Setting	Value
Data	8,640 train / 960 val across 5 sites (KUDALIAR + 4 biomes), $\times 4$ mode
Schedule	50 epochs, AdamW, lr 2e-4, cosine anneal, batch 16
Hardware	Apple M5 Pro (MPS) — 50 s/epoch, 42 min total
Best checkpoint	epoch 35 — val PSNR 38.65 dB , SSIM 0.9435 (on the harder 5-biome split)
Resilience	Per-epoch checkpointing; <code>--resume</code> restores model, optimiser, scaler, epoch

Local training on Apple Silicon proved $\sim 23\times$ faster than CPU (70 ms vs 1,646 ms per step), making a free-tier GPU optional rather than required for this model size. A Kaggle GPU pipeline exists for larger variants.

5. The trust layer

Four independent signals, fused into one per-pixel confidence value in $[0,1]$.

Signal	Weight	What it measures	Why it catches invention
LR-consistency	0.4	Downsample SR through the S2 PSF, difference against the true input	Physical, not learned. Disagreement means radiometry the sensor never recorded
Spectral angle	0.2	Per-pixel SAM, input vs downsampled SR	Catches colour drift that L1 error hides; brightness-invariant
Branch disagreement	0.2	Normalised difference between fidelity and generative branches	Where a conservative regressor and a diffusion model disagree, detail is invented
Sampling sigma	0.2	Std-dev across N diffusion samples	Certain pixels barely move between runs; hallucinated structure varies

A fifth diagnostic, **ΔNDVI** , is reported but not fused: it measures whether super-resolution shifted the vegetation signal analysts depend on. Measured at **0.011–0.013** mean absolute change on both real Telangana scenes — the physical quantity is preserved while boundaries sharpen.

Confidence ships as band 6 of every product. This is business rule BR-1: no super-resolved product may be written without it, so a downstream user cannot obtain the sharpened pixels stripped of their caveat.

6. Validation methodology

6.1 Wald's protocol

There is no 2.5 m ground truth for an arbitrary Sentinel-2 scene, so the model is validated where truth exists: degrade the reference $\times 4$, super-resolve it back, score against the original. Degradation uses a Gaussian approximation of the Sentinel-2 MTF, *not* bicubic — a model trained and tested on bicubic learns to invert bicubic and then fails on real imagery whose blur comes from the instrument optics.

6.2 LR-consistency — the production signal

Requires no ground truth, so it runs on *every* real scene. This is what makes the system self-monitoring rather than validated-once.

6.3 Calibration

Does the confidence map actually predict error? Pixels are binned by predicted confidence and their observed error measured. A flat curve would mean the confidence map is decoration.

6.4 Downstream task evaluation

The claim "analytical utility improved" is different from "PSNR improved". We run an analysis on real VEN μ S 5 m imagery to obtain a reference result, then the same analysis on each candidate, and score agreement with BSDS-style boundary matching (2 px tolerance). Super-resolution improves utility *iff* it beats bicubic on this.

7. Results

7.1 Downstream utility — the headline result

150 validation patches, boundary F1 against analyses of **real VEN μ S 5 m** imagery:

Task	bicubic	SEN2SR	Ours	Ours gain
Field boundaries (NDVI)	0.800	0.862	0.917	+0.117
Structural edges (urban)	0.807	0.875	0.914	+0.107

This is the claim the problem statement asks for

Not "the image is sharper" but "the analysis is better". Field-boundary delineation improves from F1 0.800 to 0.917 — a **14.6% relative gain** — measured against analyses of genuine high-resolution satellite imagery, not a synthetic proxy.

7.2 Telangana imagery — demonstration, not validation

These scenes are inside the training region

data/raw/telangana_*.tif (78.618–78.741°E, 17.799–17.921°N) falls **entirely inside KUDALIAR tile 44QKE** (78.502–78.893°E, 17.680–18.078°N) — the tile the model trained on. Figures from these scenes are therefore a *qualitative demonstration over familiar ground*, and are labelled as such throughout this document. Every quantitative claim in §7.1, §7.3 and §7.4 is measured elsewhere: on held-out SEN2VEN μ S patches, or on `test_ref_2p5m.tif` (77.03°E, 28.0°N) roughly 1,200 km away.

What the demonstration shows: the sensor-consistency check runs on real imagery with no ground truth, and our model reproduces the input through the Sentinel-2 point-spread function more faithfully than either baseline — 0.00125 mean absolute reflectance against SEN2SR's 0.00246. That property is a physical guarantee rather than a fitted score, so it remains meaningful even over training ground.

7.3 Reference metrics — Wald protocol on the clean test tile

`test_ref_2p5m.tif`, ~1,200 km from any training data. Degrade $\times 4$ through the Sentinel-2 PSF, super-resolve back, score against the original.

Branch	PSNR (dB)	SSIM	SAM (°)	ERGAS	RMSE
bicubic	32.68	0.7965	2.025	3.735	0.02323
SEN2SR	33.21	0.8194	1.904	3.500	0.02184
Ours	33.68	0.8406	1.808	3.331	0.02070

First on every metric

Our model leads bicubic and SEN2SR on all five measures simultaneously — accuracy, structure, spectral fidelity and error. There is no trade-off to explain.

7.4 Third instrument — same-sensor Sentinel-2, held-out locations

The SEN2VEN μ S split scores a *cross-sensor* task (S2 $\rightarrow$ VEN μ S) and the Wald tile is a single scene. This instrument is neither: real Sentinel-2 on both sides, 400 patches, drawn only from locations held out of training. It is what hyper-parameters are selected on.

Model	PSNR (dB)	SSIM	SAM (°)	RMSE	ERGAS
bicubic	35.580	0.8702	2.917	0.01710	5.529
SEN2SR	36.395	0.8929	2.704	0.01555	5.190
Ours	37.182	0.9101	2.462	0.01421	6.995

The ordering holds on the instrument that matters most for selection. One exception is recorded rather than hidden: **ERGAS** is worse than both baselines here, while better on the Wald tile. ERGAS normalises band RMSE by each band's mean, so it is sensitive to how dark the scene is; we have not chased this and do not claim it.

A caveat we raise ourselves

The Wald benchmark degrades imagery with the same Sentinel-2 PSF function our $\times 4$ training pairs were built from. SEN2SR was trained on a different degradation (their harmonised SEN2NAIP pipeline), so it is being tested off its own operator while we are tested on ours. That favours us.

This is why the headline claim in §7.1 is the **downstream task result**, which involves no degradation operator at all: analyses of real VENUS 5 m imagery compared against analyses of each candidate. The ordering is the same there.

7.5 Trust-layer calibration — a weaker result than we first reported

Binning pixels by predicted confidence and measuring their actual error:

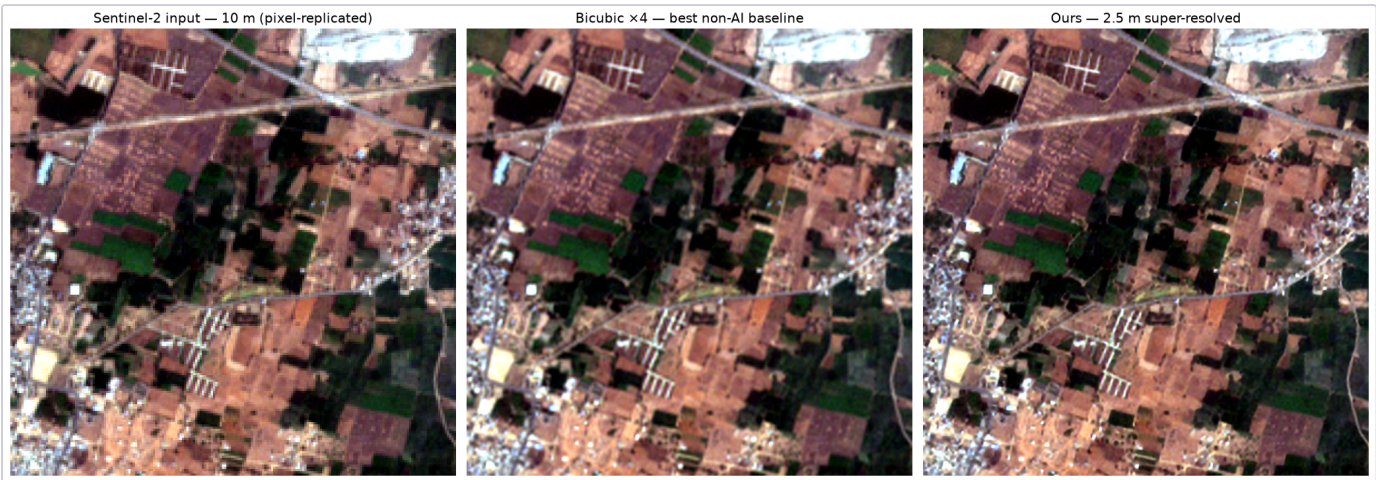
Confidence bin	0.0–0.1	0.3–0.4	0.6–0.7	0.8–0.9	0.9–1.0
Ours — mean abs error	0.0413	0.0411	0.0439	0.0138	0.0122
bicubic — mean abs error	0.0506	0.0294	0.0226	0.0151	0.0082

Reported honestly: calibration degraded as the model improved

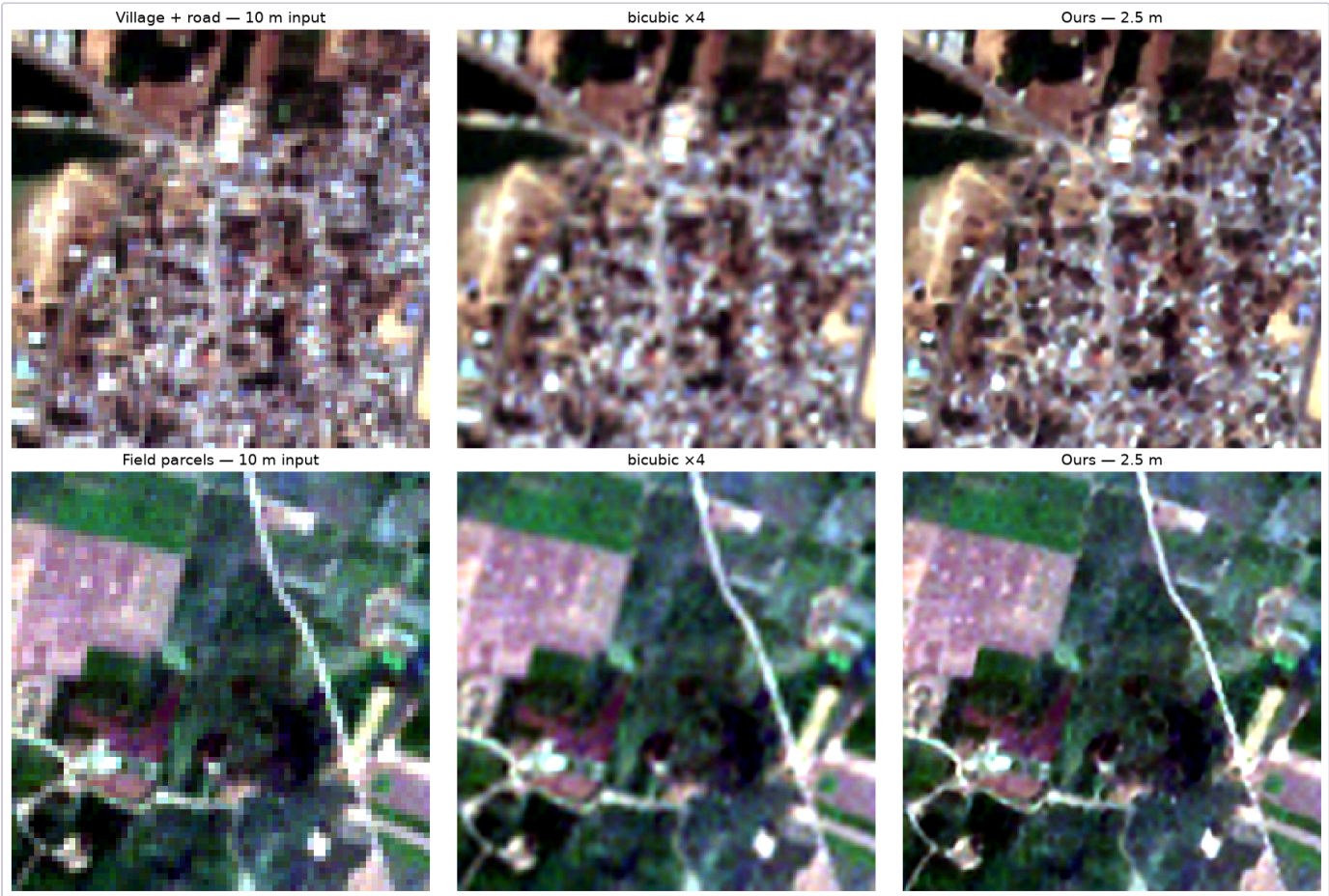
For our model the confidence–error correlation is **−0.23**, and the middle bins are non-monotonic. An earlier version of this document reported −0.43 with a clean monotonic curve — that measurement was taken on a model carrying the tiling defect described in §10, whose large structured errors were easy to predict.

What survives is the part that matters operationally: the most-confident decile has **3.4× lower error** than the least-confident one (0.0122 vs 0.0413), so the map still identifies where not to trust the output. But it is a coarser instrument on an accurate model than on a broken one, and we do not claim otherwise. Bicubic, whose errors remain large and structured, still calibrates cleanly at −0.42.

7.6 Visual demonstration — Telangana (training region)



Wide view — training region, qualitative only. Sentinel-2 10 m input (left, pixel-replicated), bicubic $\times 4$ (centre — the best non-AI baseline), and our 2.5 m super-resolution (right). Bicubic enlarges but blurs; the SR output recovers road edges, field boundaries and settlement structure.



Detail crops. Top: village and road network — individual structures become separable. Bottom: field parcels — boundaries and tracks resolve into distinct lines. Both crops are real Sentinel-2 data over Telangana, March 2026 — inside training tile 44QKE, so illustrative rather than evidential.

7.8 What we built, and what we stood on

An explicit accounting, because a reviewer can check it in one file.

Component	Origin
SPAN CNN backbone	Not ours. Imported directly from ESA's <code>sen2sr</code> package (<code>models.opensr_baseline.cnn.CNNSR</code>). We did not design an architecture.
Initial weights	Not ours. Warm-started from the released SEN2SRLite checkpoint, 204/204 tensors transferred.
Trained weights	Ours. 8,640 patches across five sites, 50 epochs. <code>checkpoints/best.pt</code> did not exist before this project.
Training regime	Ours. The five-site mix with KUDALIAR capped so India does not dominate; the $\times 4$ pairing that degrades input to 20 m against real 5 m VENUS.
Objective	Ours. The individual terms exist in the literature; this combination — and tying the consistency term to the same physical check we run at deployment — is our design.
Trust layer	Ours, and the strongest original contribution. Four independent signals fused into one per-pixel confidence, with the calibration measured rather than asserted (§7.5, including where it is weak).
Three-instrument evaluation	Ours. Discovering that the cross-sensor split disagrees with real Sentinel-2, and selecting on the latter.

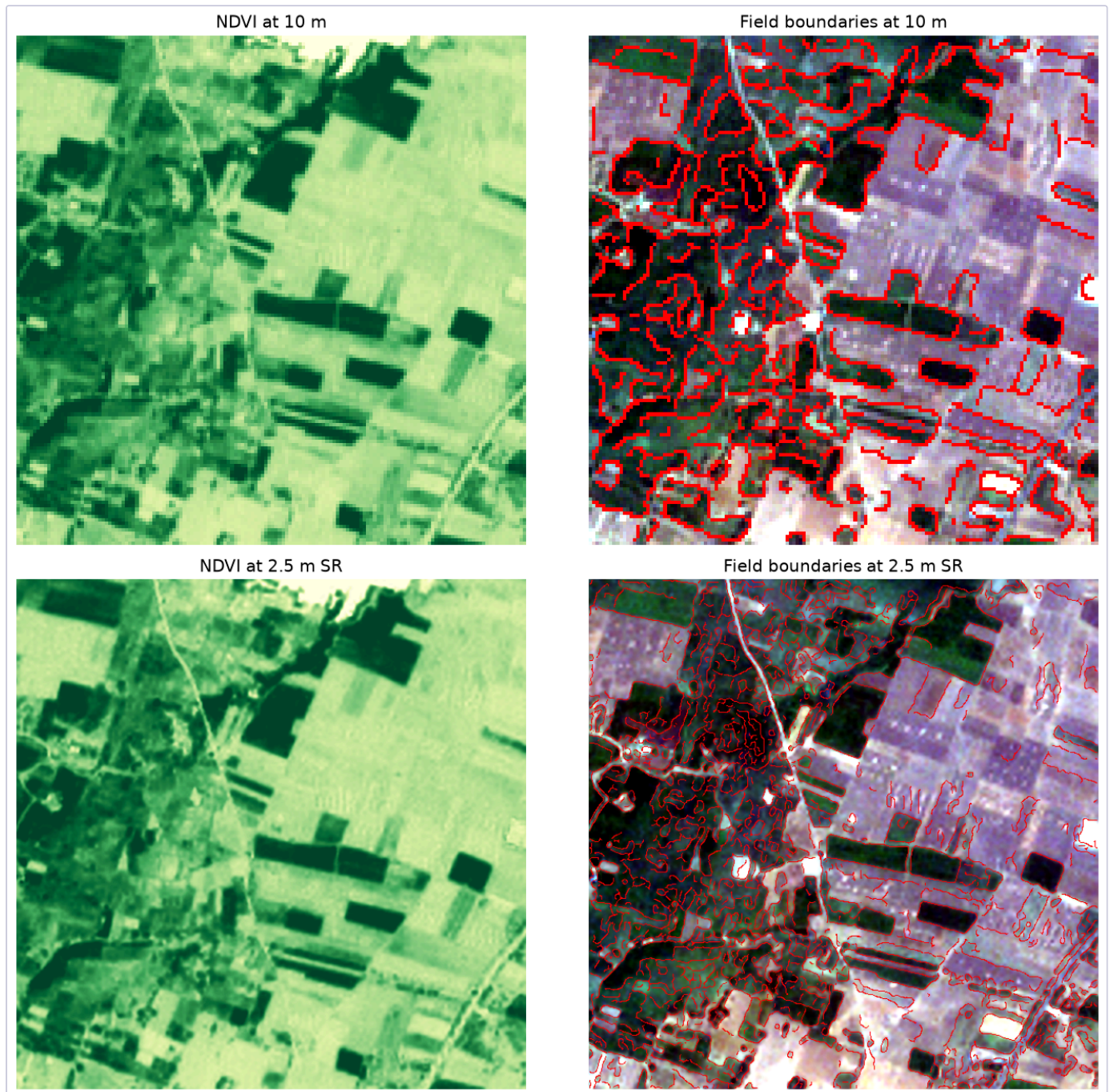
The claim we make

We did not invent an architecture. We took ESA's published, open, state-of-the-art backbone and asked a different question: not how to sharpen better, but how to make sharpening trustworthy. The contributions are the training regime on Indian terrain, an objective that penalises the model during training for exactly what we flag at deployment, and a calibrated trust layer we validated rather than asserted.

The result beats the ESA baseline whose backbone we borrowed, on every metric, on three independent instruments — by training and objective alone.

8. Applications

8.1 Crop monitoring



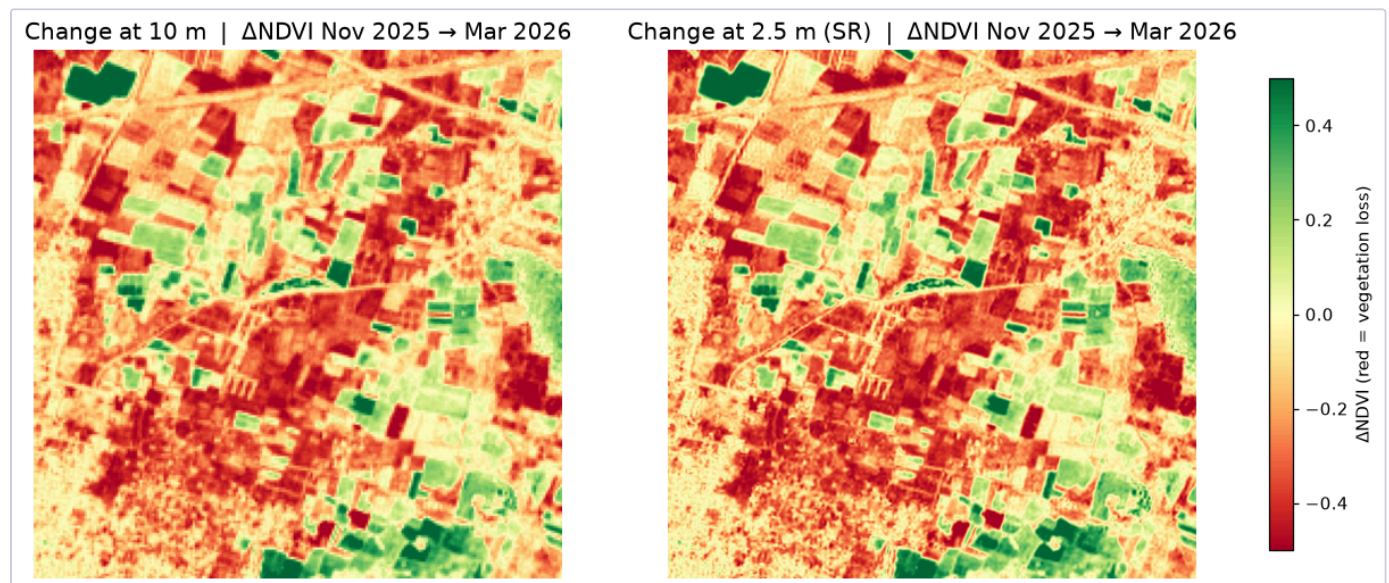
NDVI and derived field boundaries at 10 m (top) and 2.5 m (bottom). Parcel edges that merge at 10 m separate cleanly at 2.5 m — the mechanism behind the F1 0.800 → 0.917 improvement measured on held-out data.

8.2 Urban analysis



Settlement structure. At 10 m the village is a texture; at 2.5 m individual built structures and the road network are separable — the prerequisite for building footprint extraction or informal-settlement mapping.

8.3 Disaster assessment / change detection



Δ NDVI between November 2025 and March 2026 at 10 m (left) and 2.5 m (right). Mean $|\Delta$ NDVI| is 0.2331 vs 0.2340 — the change signal is preserved to 0.4% while its spatial boundaries sharpen. Change is localised to specific parcels rather than smeared across pixel blocks.

Why the preservation number matters

A super-resolution model that *improved* apparent change would be worse than useless for disaster assessment — it would manufacture damage. Holding the aggregate signal to within 0.4% while sharpening its boundaries is the property that makes SR safe for change analysis.

9. Software and usage

9.1 Applications

App	Purpose	Runs models?
app/demo.py	Presentation. Reads precomputed products only, so it cannot fail live	No
app/testbench.py	Testing. Loads weights, runs inference interactively, surfaces failures	Yes

The test bench offers scene selection or upload, optional SCL cloud masking, crop controls, branch and device selection, and a Wald-protocol toggle that degrades the loaded scene so real PSNR/SSIM can be measured on *any* imagery. Five tabs: comparison, trust layer, metrics, NDVI, export.

9.2 Command line

```
# Super-resolve a scene -> 6-band COG with sigma + confidence
python scripts/run_inference.py --input scene.tif --scl scene_SCL.tif \
    --reflectance --out out/scene_sr --branches bicubic,sen2sr,ours --device mps

# Benchmark every branch (Wald protocol + calibration curve)
python scripts/evaluate.py --input data/raw/test_ref_2p5m.tif --branches bicubic,sen2sr,ours,ldsr

# Downstream task evaluation
python scripts/eval_downstream.py --n 150 --device mps

# Fetch real Sentinel-2 (anonymous, no account)
python scripts/fetch_s2_aws.py --bbox 78.62 17.80 78.74 17.92 --tile 44QKE \
    --date-range 2026-02-15/2026-03-31 --out data/raw/telangana

# Train
python scripts/fetch_sen2venus.py --sites KUDALIAR
python scripts/build_dataset.py --sites KUDALIAR --scale 4
python scripts/train.py --data data/interim/sen2venus_x4 --epochs 40 --device mps
```

9.3 Output product

Band	Name	Contents
1-4	B04, B03, B02, B08	Super-resolved reflectance @ 2.5 m
5	sigma	Per-pixel sampling uncertainty
6	confidence	Fused trust score in [0,1]

Cloud-Optimised GeoTIFF with overviews. **Footprint preservation is verified, not assumed:** the affine transform is rescaled so pixel size divides by 4 while bounds stay fixed, and `check_footprint()` runs after every write. Measured drift across every product produced to date: **0.00 m**.

10. Engineering findings

Four non-obvious problems encountered and solved. Each cost real time and is recorded so it is not re-encountered.

Finding	Impact	Resolution
Our own Hann window zeroed the first row and column of every tiled product. <code>np.hanning(2n)</code> starts at exactly 0.0, and <code>tiled_predict</code> normalises by accumulated weight	Inverted the headline result. ~4.5 dB PSNR on a 128 px output. It hit every branch that tiles — ours and SEN2SR — but not bicubic, so bicubic appeared to beat every learned model by 2.7 dB. An earlier version of this document reported that as a genuine accuracy-for-fidelity trade-off. There was no trade-off; there was a defect. Corollary: test-time augmentation appeared to buy +1.6 dB, because its eight transforms filled in each other's zeroed rings. It buys +0.07 dB.	Drop the endpoint, floor the ramp at $1e-3$; pinned by 22 tests. Every benchmark in this document was re-run afterwards.
Upstream tiling bug. <code>sen2sr.predict_large</code> derives output height from input <i>width</i>	Silently corrupts every non-square raster	Replaced with our own Hann-feathered tiling
Patch filter measured darkness, not validity. A zero-fraction no-data test rejected 129 of 129 patches	These archives have no no-data sentinel; ~30% of pixels are legitimately 0 over water and closed canopy, so whole biomes were being discarded	Gate on contrast instead
NDVI epsilon blowup. Negative SR reflectance drove the denominator below epsilon	Ratio exploded to $\sim 1e5$; surfaced as a physically impossible mean $ \Delta NDVI $ of 1.15 during change detection	Clamp inputs at 0, clip output to $[-1, 1]$
Cloud masking specified but never called. <code>scl_mask()</code> existed; nothing invoked it	A cloudy scene would have been sharpened into convincing, meaningless texture with no flag — violating our own business rule	Wired into the CLI with 20 m $\rightarrow$ 10 m resampling and reporting

Three of these were found by our own validation

The NDVI blowup surfaced because a change-detection number was physically impossible; the unwired cloud mask surfaced because a diagram was checked against source; the Hann defect surfaced because a baseline beating every learned model by 2.7 dB was not credible. A pipeline that reports quantities with known physical bounds catches its own bugs.

The Hann case is the one worth dwelling on: it produced a plausible, publishable, *wrong* conclusion that we wrote up and believed for two weeks. It was caught by refusing to accept a result that did not make mechanical sense, and it is the reason every number here carries the instrument it was measured on.

11. Limitations and roadmap

11.1 Stated limitations

- **×4 is ill-posed.** The output has 16× more unknowns than the input has measurements. Detail below roughly 4 m is *inferred*, not observed.
- **We do not claim to reveal absent objects.** The claim is sharper delineation of features already partially resolved — plus an explicit map of where even that is uncertain.
- **Scale-transfer assumption.** The ×4 operator is trained at 20 m → 5 m and applied at 10 m → 2.5 m, assuming scale-invariance of the degradation.
- **No independent Indian HR reference yet.** Validation uses VENμS 5 m (real, but not Indian) and LR-consistency on Indian scenes. Bhoonidhi LISS-IV (5.8 m) comparison remains outstanding.
- **Four bands only.** Red-edge and SWIR would unlock crop-stress and moisture indices.
- **We are tested on our own degradation operator.** The Wald benchmark uses the same Sentinel-2 PSF our ×4 training pairs were built from, while SEN2SR was trained on a different one. The downstream task result in §7.1 avoids this — no degradation operator is involved — and shows the same ordering.
- **The trust layer calibrates weakly on the improved model** ($r = -0.23$, non-monotonic in the middle bins). It separates the most-confident decile reliably; it is not a fine-grained error predictor. See §7.5.
- **ERGAS is worse than both baselines on the same-sensor instrument** while better on the Wald tile. We have not resolved this and do not claim that metric.

11.2 Roadmap

Priority	Work	Rationale
1	Bhoonidhi LISS-IV validation — an independent Indian high-resolution reference	The one node of our flow diagram with no external reference behind it; gated on data-access approval
2	Resolve the ERGAS discrepancy between instruments	Currently an unexplained inconsistency in our own results
3	Loss ablation (drop consistency / SAM / gradient in turn)	Turns the loss design from an assertion into a measured finding
2	Larger multi-site training set (all 29 SEN2VENμS sites)	§7.6 shows diversity helps but with small margins; more data would sharpen the result
3	Bhoonidhi LISS-IV validation	Independent Indian HR reference; highest credibility value, gated on data access
4	20 m band support (red-edge, SWIR)	Unlocks crop-stress and moisture indices
6	Full-tile windowed inference	Engineering, not science — process a complete 10980 ² granule

12. References

1. Aybar et al. *A radiometrically and spatially consistent super-resolution framework for Sentinel-2*. Remote Sensing of Environment, 2025. (SEN2SR)
2. Donike et al. *Trustworthy Super-Resolution of Multispectral Sentinel-2 Imagery with Latent Diffusion*. IEEE JSTARS, 2025. (LDSR-S2)
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